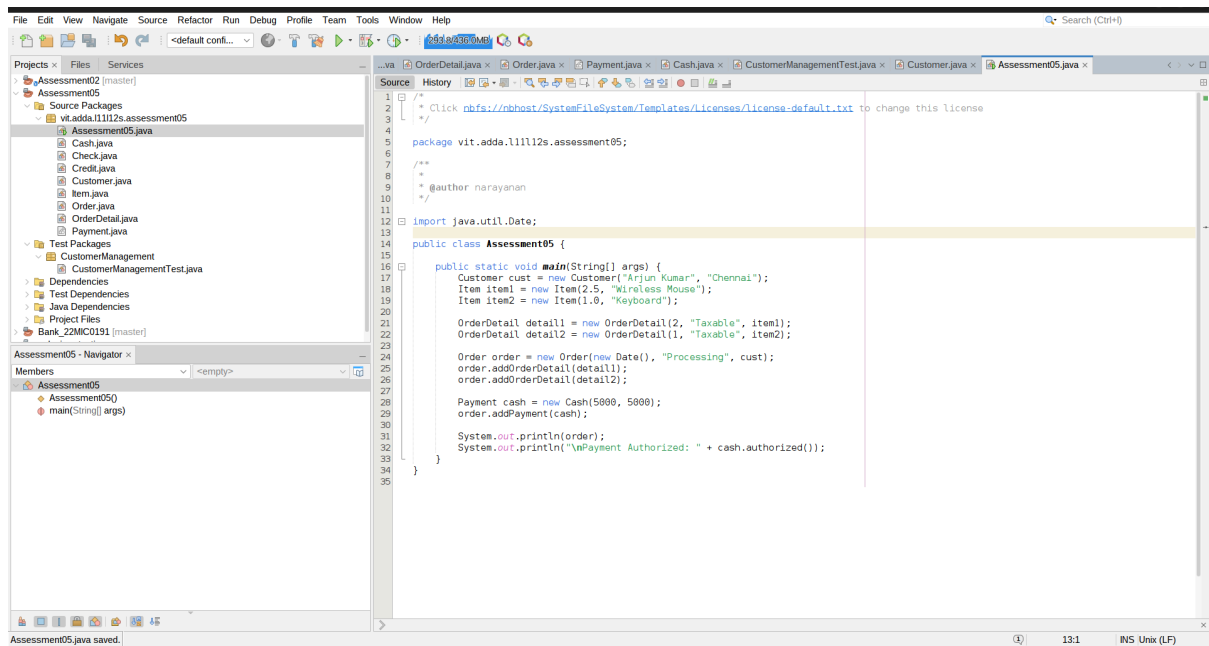


 VIT[®] Vellore Institute of Technology (Deemed to be University under section 3 of the UGC Act, 1956)	Project	
Fall Semester 2025-2026		
Course code: CSI3025	Course Title :Application Development and Deployment Architecture	Slot : F2
	Course type: ETH + ELA	Slot : L11+L12
Class No:4948	School: SCOPE	Venue: SJT 318
Emp. ID: 20774	Faculty Name: SUDHAKAR. P	Code : B

NAME: LAKSHMI NARAYANAN A
REG.NO: 22MIC0191
DATE: 16/10/2025

Main class



```

1  /**
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   */
4
5  package vit.adda.l11l12s.assessment05;
6
7
8  /**
9   *
10  * @author narayanan
11  */
12
13  import java.util.Date;
14
15  public class Assessment05 {
16
17      public static void main(String[] args) {
18          Customer cust = new Customer("Arjun Kumar", "Chennai");
19          Item item1 = new Item(2.5, "Wireless Mouse");
20          Item item2 = new Item(1.0, "Keyboard");
21
22          OrderDetail detail1 = new OrderDetail(2, "Taxable", item1);
23          OrderDetail detail2 = new OrderDetail(1, "Taxable", item2);
24
25          Order order = new Order(new Date(), "Processing", cust);
26          order.addOrderDetail(detail1);
27          order.addOrderDetail(detail2);
28
29          Payment cash = new Cash(5000, 5000);
30          order.addPayment(cash);
31
32          System.out.println(order);
33          System.out.println("\nPayment Authorized: " + cash.authorized());
34      }
35  }

```

Customer class

The screenshot shows an IDE with the following components:

- Projects:** A tree view on the left showing the project structure. The 'Customer' class is highlighted under the 'Test Packages' section.
- Members:** A panel below the Projects view showing the members of the 'Customer' class: `Customer(String name, String address)`, `getAddress(): String`, `getName(): String`, `toString(): String + Object`, `address: String`, and `name: String`.
- Source:** The main editor window displaying the source code of the 'Customer' class. The code is as follows:

```
1  /**  
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license  
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template  
4   */  
5  package vit.adda.l11112s.assessment05;  
6  
7  /**  
8   *  
9   * @author narayanan  
10  */  
11 public class Customer {  
12     private String name;  
13     private String address;  
14  
15     public Customer(String name, String address) {  
16         this.name = name;  
17         this.address = address;  
18     }  
19     public String getName() { return name; }  
20     public String getAddress() { return address; }  
21  
22     @Override  
23     public String toString() {  
24         return "Customer Name: " + name + ", Address: " + address;  
25     }  
26 }  
27  
28
```
- Test Results:** A panel at the bottom showing the test results for the 'Customer' class, indicating that the test passed.

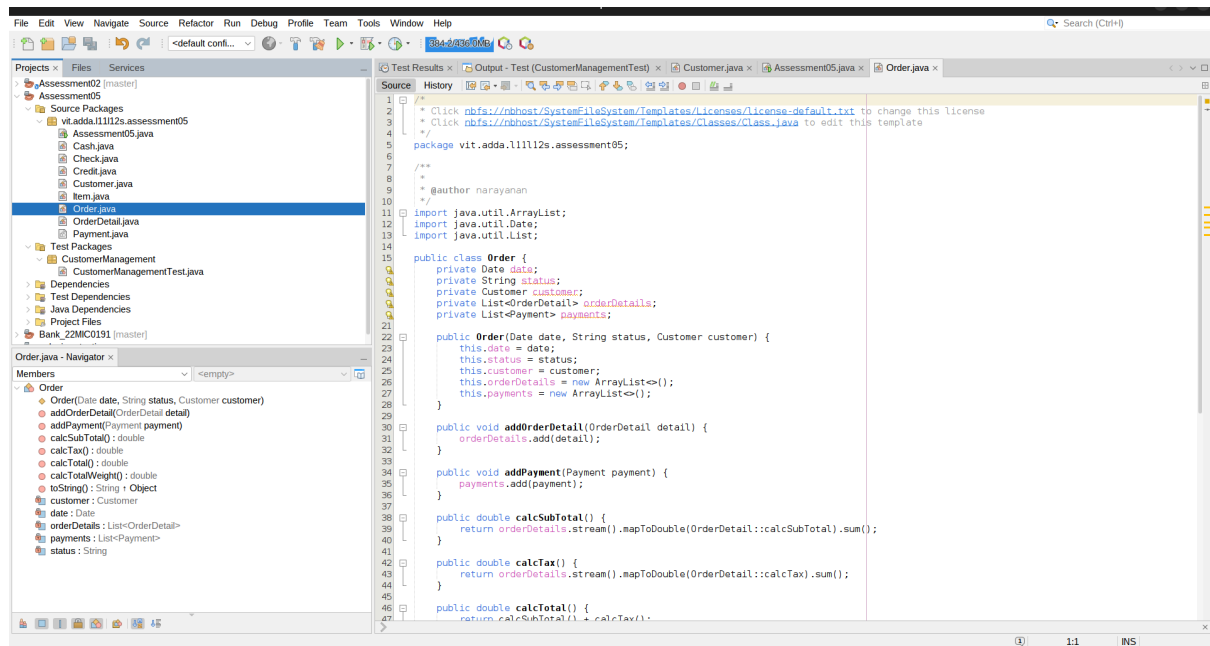
Item class

The screenshot shows an IDE with the following components:

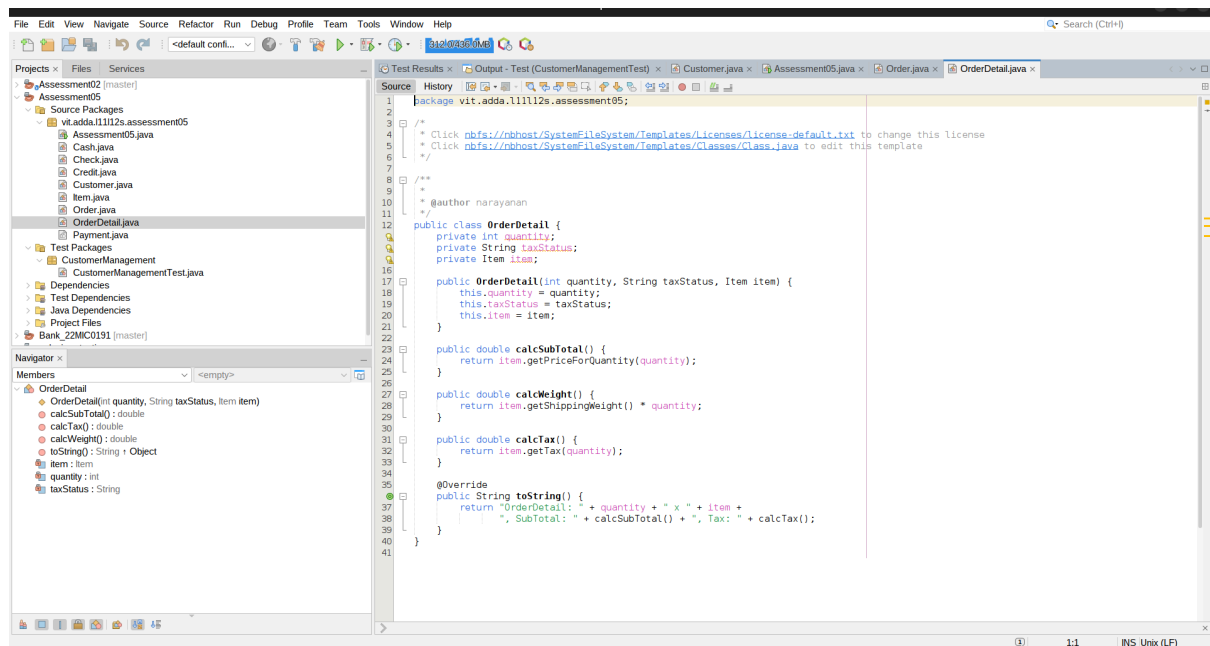
- Projects:** A tree view on the left showing the project structure. The 'Item' class is highlighted under the 'Test Packages' section.
- Members:** A panel below the Projects view showing the members of the 'Item' class: `Item(double shippingWeight, String description)`, `getPriceForQuantity(int quantity): double`, `getShippingWeight(): double`, `getTax(int quantity): double`, `inStock(): boolean`, `description: String`, and `shippingWeight: double`.
- Source:** The main editor window displaying the source code of the 'Item' class. The code is as follows:

```
1  package vit.adda.l11112s.assessment05;  
2  
3  /**  
4   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license  
5   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template  
6   */  
7  
8  /**  
9   *  
10  * @author narayanan  
11  */  
12 public class Item {  
13     private double shippingWeight;  
14     private String description;  
15  
16     public Item(double shippingWeight, String description) {  
17         this.shippingWeight = shippingWeight;  
18         this.description = description;  
19     }  
20  
21     public double getPriceForQuantity(int quantity) {  
22         return quantity * 100; // Example price calculation  
23     }  
24  
25     public double getTax(int quantity) {  
26         return getPriceForQuantity(quantity) * 0.05; // 5% tax  
27     }  
28     public boolean inStock() {  
29         return true;  
30     }  
31  
32     public double getShippingWeight() { return shippingWeight; }  
33 }  
34  
35
```
- Test Results:** A panel at the bottom showing the test results for the 'Item' class, indicating that the test passed.

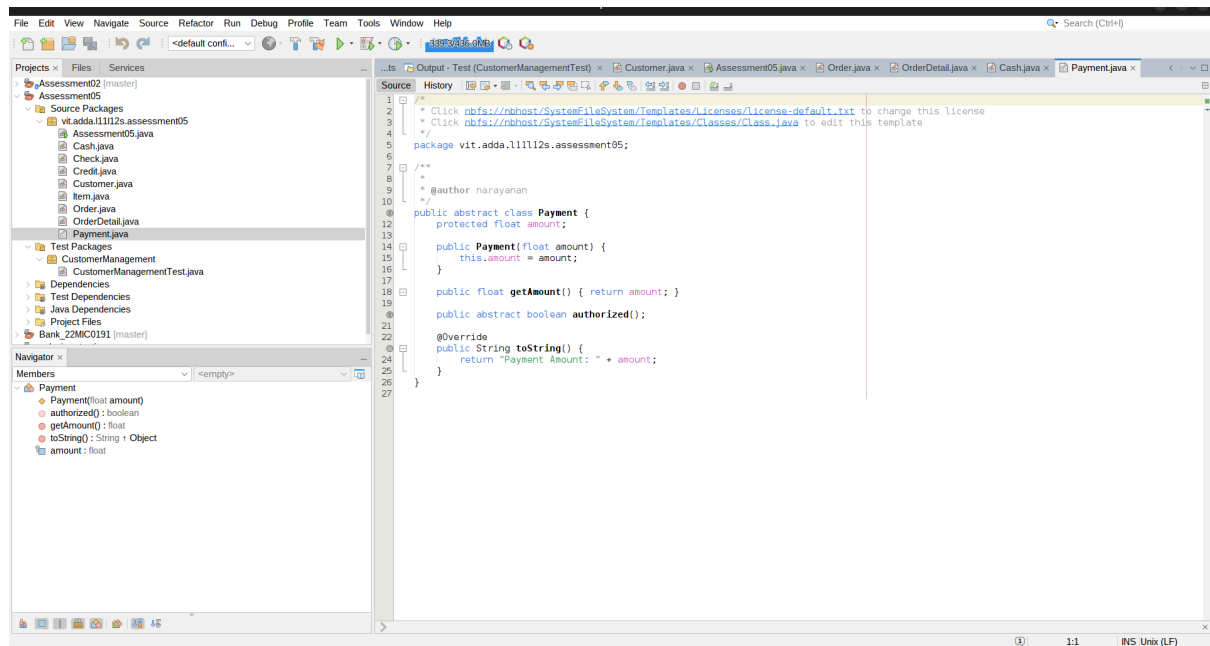
Order class



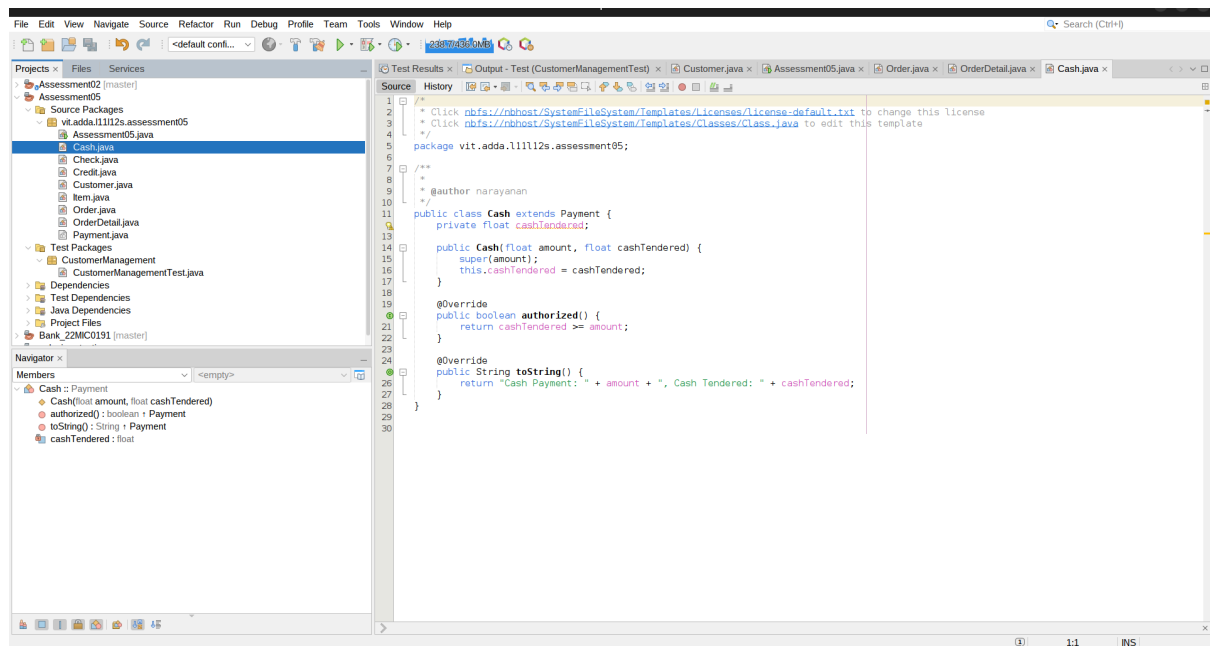
OrderDetail class



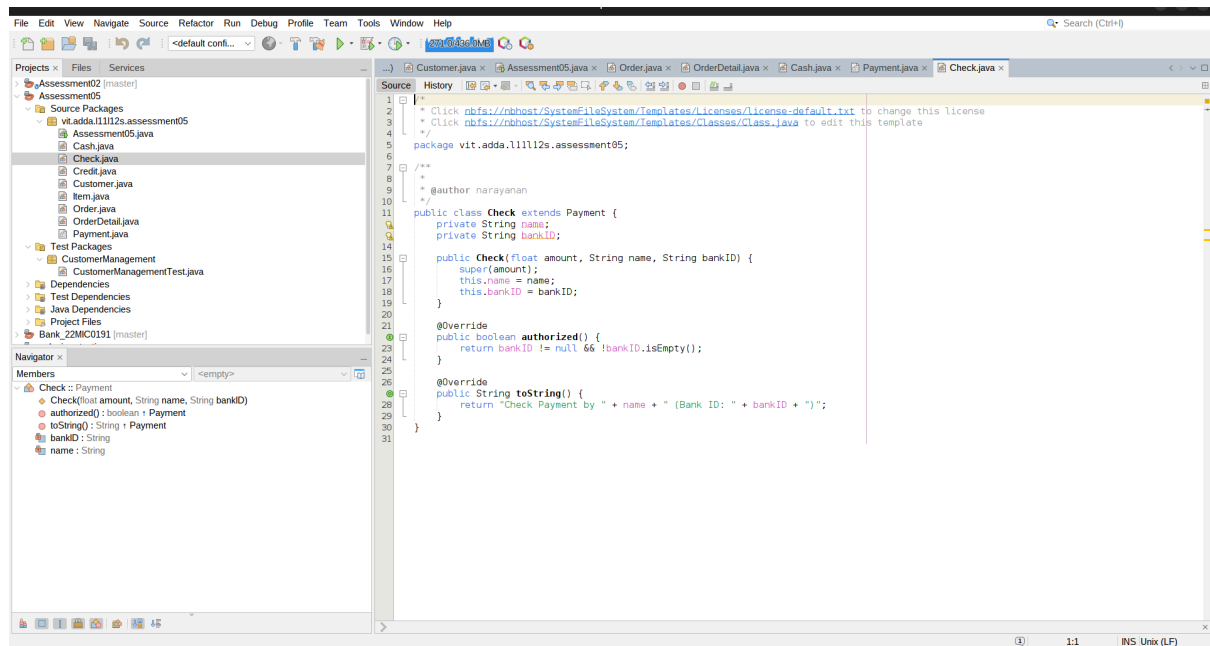
Payment class



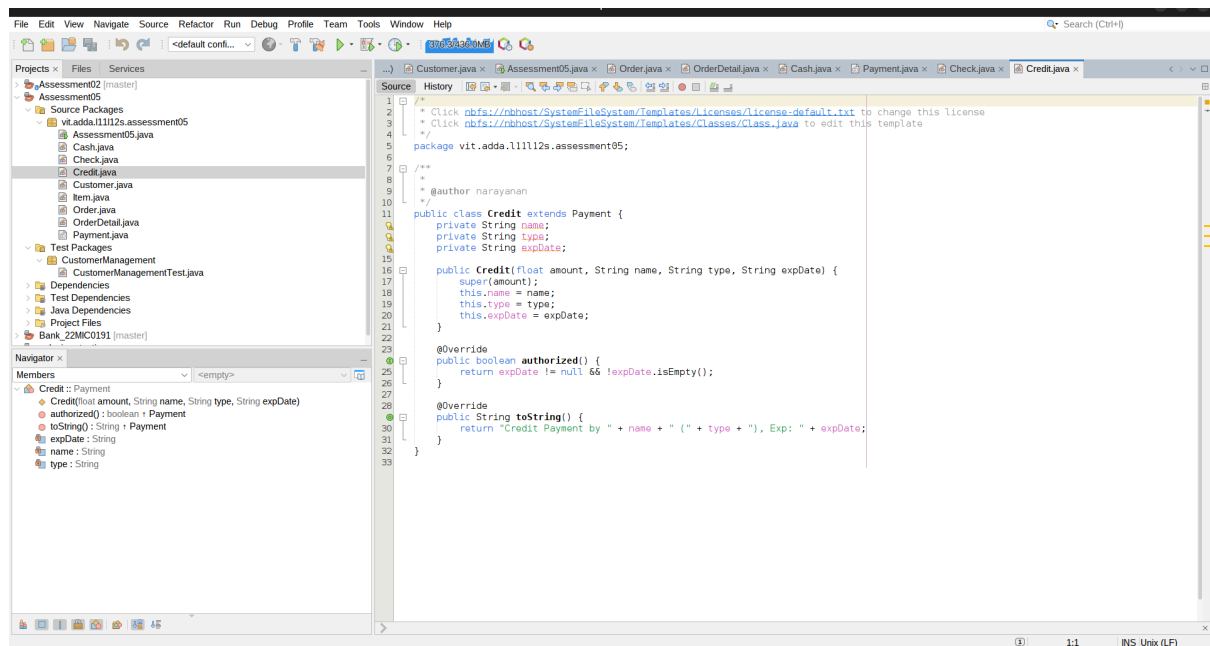
Cash class



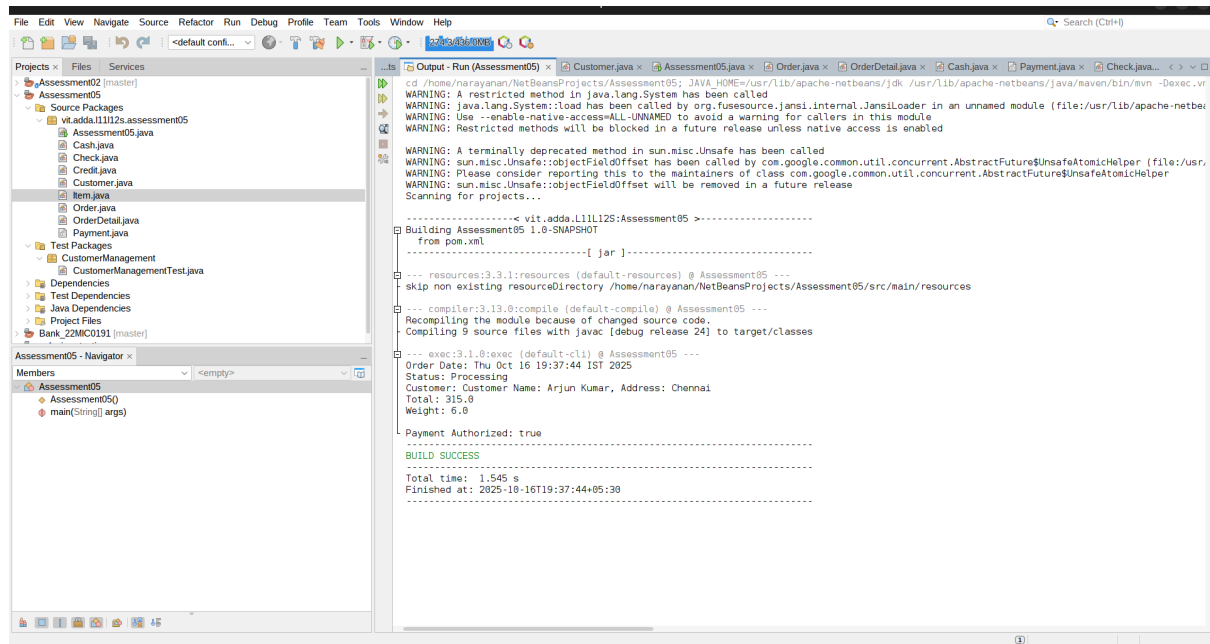
Check class



Credit class



OUTPUT



The screenshot shows the NetBeans IDE interface. The 'Output - Run (Assessment05)' window is active, displaying the following text:

```
cd /home/narayanan/NetBeansProjects/Assessment05; JAVA_HOME=/usr/lib/apache-netbeans/jdk /usr/lib/apache-netbeans/java/maven/bin/mvn -Dexec.vr
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by org.fusesource.jansi.internal.JansiLoader in an unnamed module (file:/usr/lib/apache-netbe
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.google.common.util.concurrent.AbstractFuture$UnsafeAtomicHelper (file:/usr/
WARNING: Please consider reporting this to the maintainers of class com.google.common.util.concurrent.AbstractFuture$UnsafeAtomicHelper
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
Scanning for projects...

----- vit.adda.l11112s.Assessment05 -----
Building Assessment05 1.0-SNAPSHOT
from pom.xml
-----[ jar ]-----
--- resources:3.3.1:resources (default-resources) @ Assessment05 ---
skip non existing resourceDirectory /home/narayanan/NetBeansProjects/Assessment05/src/main/resources
--- compiler:3.13.0:compile (default-compile) @ Assessment05 ---
Recompiling the module because of changed source code.
Compiling 9 source files with javac [debug release 24] to target/classes

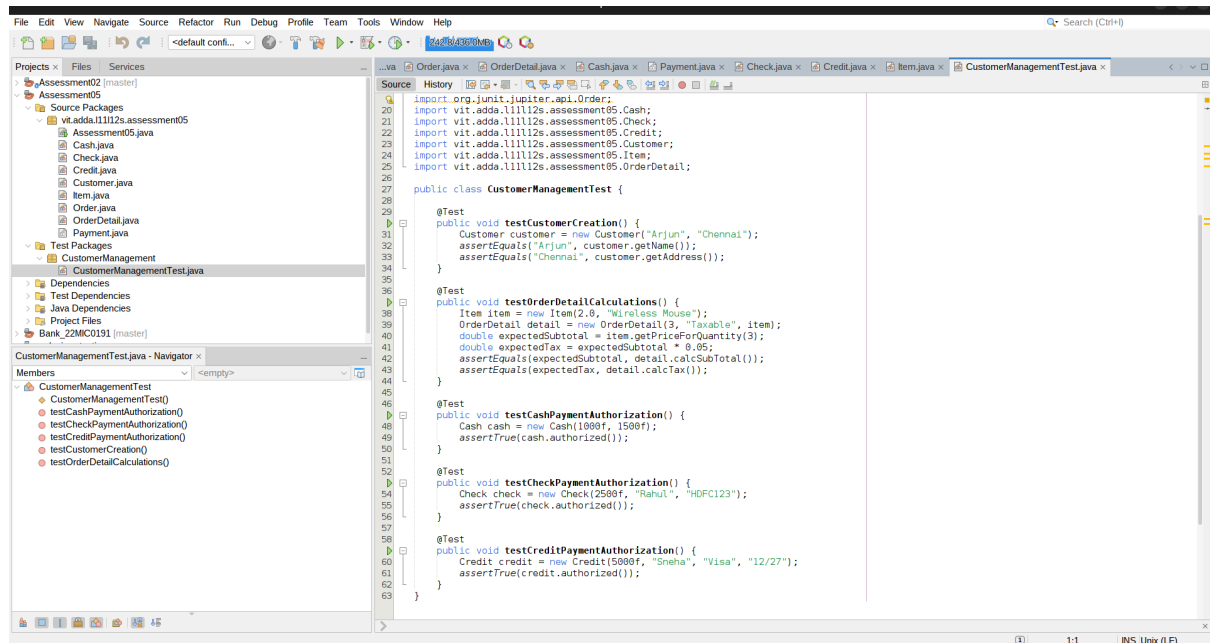
--- exec:3.1.0:exec (default-cli) @ Assessment05 ---
Order Date: Thu Oct 16 19:37:44 IST 2025
Status: Processing
Customer: Customer Name: Arjun Kumar, Address: Chennai
Total: 215.0
Weight: 6.0
Payment Authorized: true

BUILD SUCCESS

Total time: 1.545 s
Finished at: 2025-10-16T19:37:44+05:30
```

TEST

TestFile



The screenshot shows the NetBeans IDE interface. The 'CustomerManagementTest.java' file is open in the editor. The code is as follows:

```
import org.junit.jupiter.api.Order;
import vit.adda.l11112s.assessment05.Cash;
import vit.adda.l11112s.assessment05.Check;
import vit.adda.l11112s.assessment05.Credit;
import vit.adda.l11112s.assessment05.Customer;
import vit.adda.l11112s.assessment05.Item;
import vit.adda.l11112s.assessment05.OrderDetail;

public class CustomerManagementTest {

    @Test
    public void testCustomerCreation() {
        Customer customer = new Customer("Arjun", "Chennai");
        assertEquals("Arjun", customer.getName());
        assertEquals("Chennai", customer.getAddress());
    }

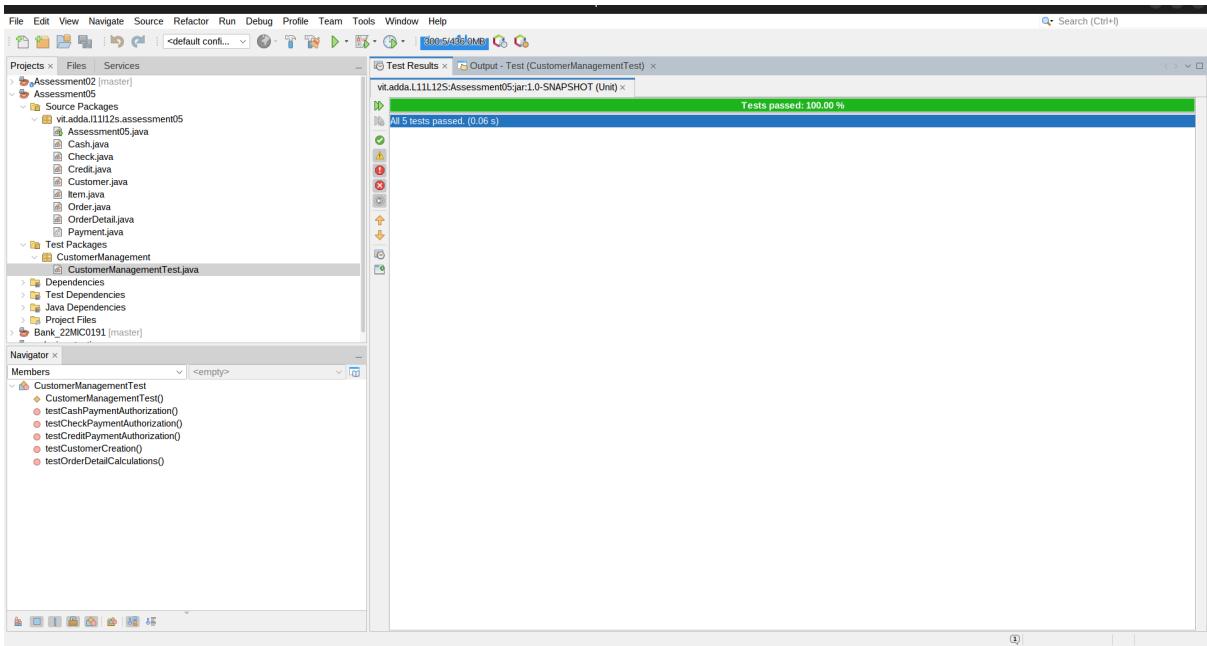
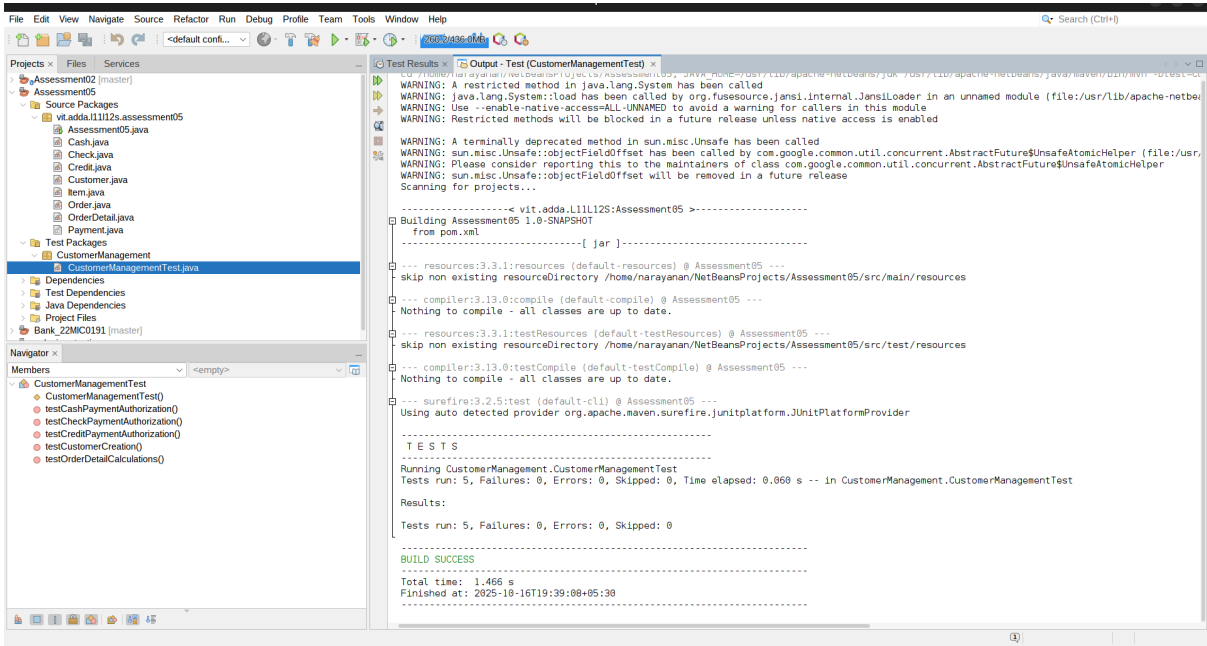
    @Test
    public void testOrderDetailCalculations() {
        Item item = new Item(2.0, "Wireless Mouse");
        OrderDetail detail = new OrderDetail(3, "Taxable", item);
        double expectedSubtotal = item.getPriceForQuantity(3);
        double expectedTax = expectedSubtotal * 0.05;
        assertEquals(expectedSubtotal, detail.calcSubTotal());
        assertEquals(expectedTax, detail.calcTax());
    }

    @Test
    public void testCashPaymentAuthorization() {
        Cash cash = new Cash(1000f, 1500f);
        assertTrue(cash.authorized());
    }

    @Test
    public void testCheckPaymentAuthorization() {
        Check check = new Check(2500f, "Rahul", "HDFC123");
        assertTrue(check.authorized());
    }

    @Test
    public void testCreditPaymentAuthorization() {
        Credit credit = new Credit(5000f, "Sneha", "Visa", "12/27");
        assertTrue(credit.authorized());
    }
}
```

Test Results



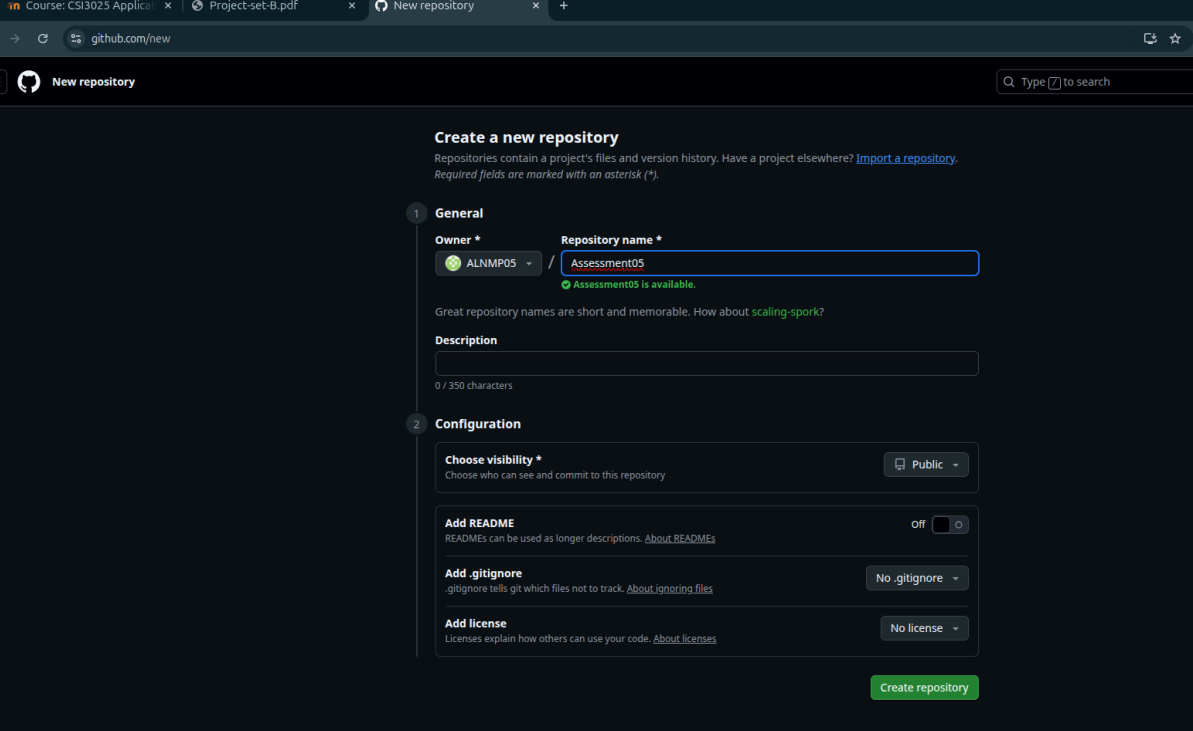
Question 3: Consider your software development team is spread across different locations. How do you manage the code repository using Github? Demonstrate your code repository with necessary screenshots and explanations for the same.

Initialising Local Repository

```
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/narayanan/NetBeansProjects/Assessment05/.git/
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$

hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/narayanan/NetBeansProjects/Assessment05/.git/
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git add .
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git commit -m "initial commit"
[master (root-commit) 7aa6e91] initial commit
28 files changed, 496 insertions(+)
create mode 100644 nbproject/project.properties
create mode 100644 pom.xml
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/Assessment05.java
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/Cash.java
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/Check.java
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/Credit.java
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/Customer.java
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/Item.java
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/Order.java
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/OrderDetail.java
create mode 100644 src/main/java/vit/adda/l11l12s/assessment05/Payment.java
create mode 100644 src/test/java/CustomerManagement/CustomerManagementTest.java
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/Assessment05.class
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/Cash.class
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/Check.class
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/Credit.class
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/Customer.class
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/Item.class
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/Order.class
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/OrderDetail.class
create mode 100644 target/classes/vit/adda/l11l12s/assessment05/Payment.class
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/CustomerManagement.CustomerManagementTest.txt
create mode 100644 target/surefire-reports/TEST-CustomerManagement.CustomerManagementTest.xml
create mode 100644 target/test-classes/CustomerManagement/CustomerManagementTest.class
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$
```

Creating remote repository in github for global access



Course: CSI3025 Application Development x Project-set-B.pdf x New repository x +

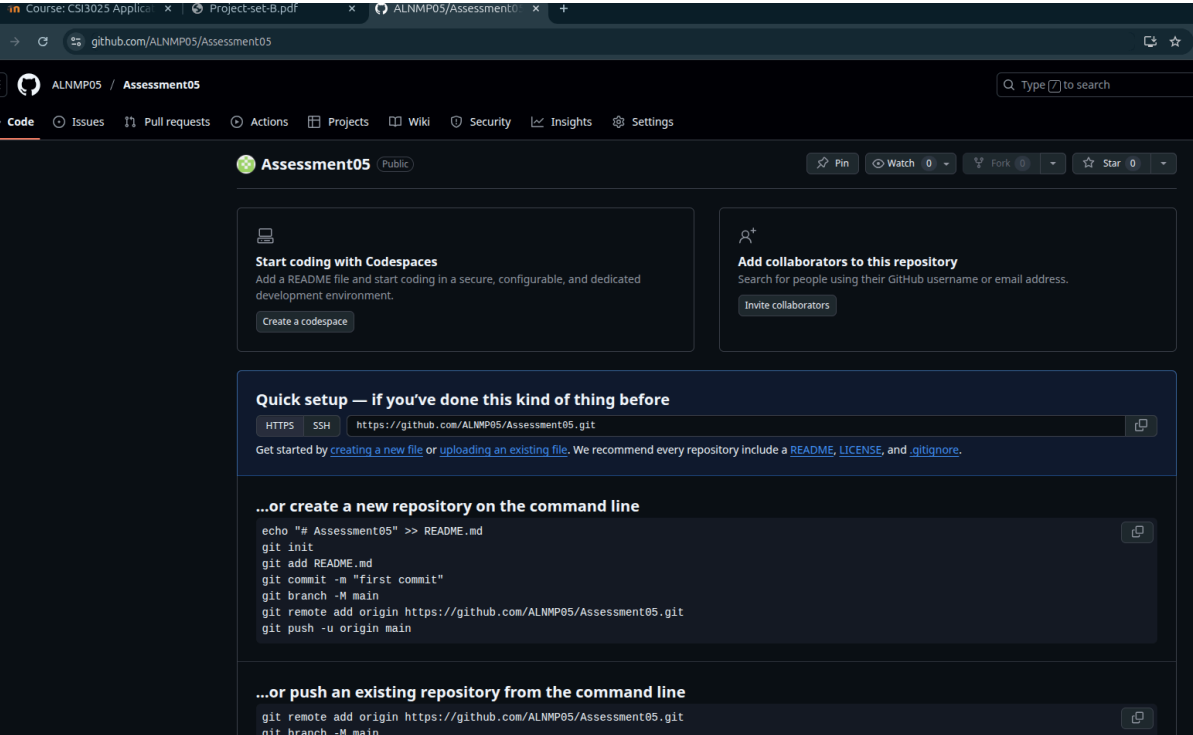
github.com/new

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

- General**
 - Owner *** ALNMP05
 - Repository name *** Assessment05
Assessment05 is available.
 - Great repository names are short and memorable. How about [scaling-spork?](#)
 - Description**
0 / 350 characters
- Configuration**
 - Choose visibility *** Public
Choose who can see and commit to this repository
 - Add README** Off
READMEs can be used as longer descriptions. [About READMEs](#)
 - Add .gitignore** No .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)
 - Add license** No license
Licenses explain how others can use your code. [About licenses](#)

Create repository



Course: CSI3025 Application Development x Project-set-B.pdf x ALNMP05/Assessment05 x +

github.com/ALNMP05/Assessment05

ALNMP05 / Assessment05

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Assessment05 Public

Pin Watch 0 Fork 0 Star 0

Start coding with Codespaces
Add a README file and start coding in a secure, configurable, and dedicated development environment.
[Create a codespace](#)

Add collaborators to this repository
Search for people using their GitHub username or email address.
[Invite collaborators](#)

Quick setup — if you've done this kind of thing before

HTTPS SSH `https://github.com/ALNMP05/Assessment05.git`

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# Assessment05" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/ALNMP05/Assessment05.git
git push -u origin main
```

...or push an existing repository from the command line

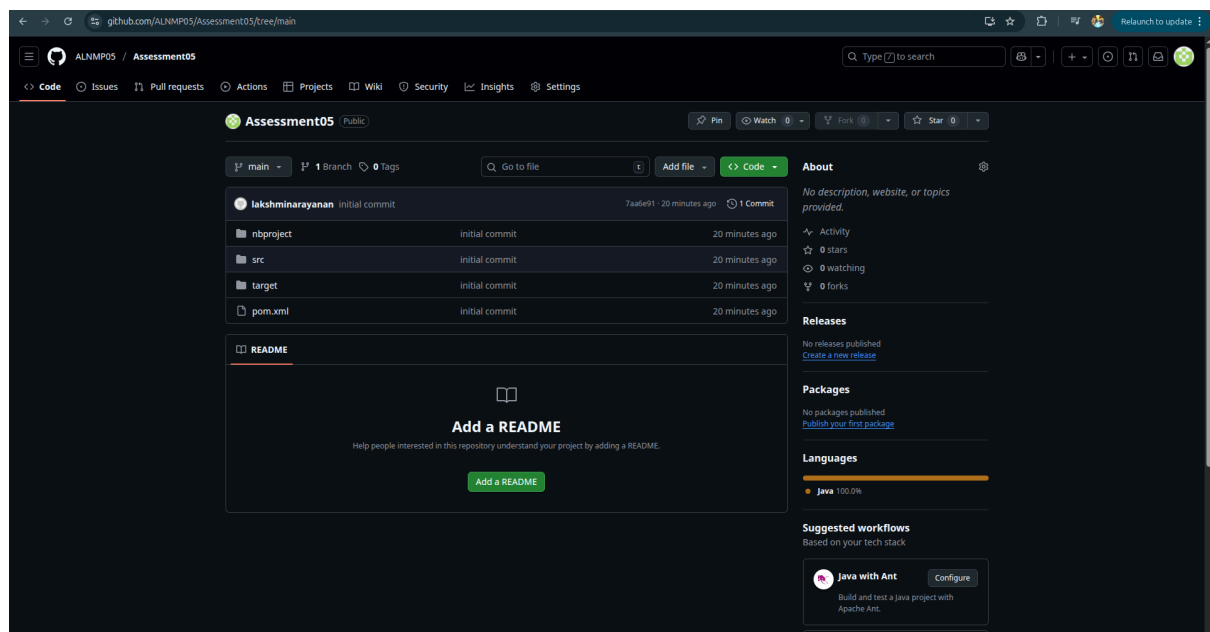
```
git remote add origin https://github.com/ALNMP05/Assessment05.git
git branch -M main
git push -u origin main
```

Pushing local repository to github

Created SSH key and added it in github for remote connection

Pushed project to github

```
create mode 100644 target/surefire-reports/CustomManagementTest.txt
create mode 100644 target/surefire-reports/TEST-CustomerManagementTest.xml
create mode 100644 target/test-classes/CustomManagementTest.class
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git remote add origin https://github.com/ALNMP05/Assessment05.git
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git branch -M main
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git push -u origin main
Username for 'https://github.com': ALNMP05
Password for 'https://ALNMP05@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/ALNMP05/Assessment05.git/'
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git push -u origin main
Username for 'https://github.com': ALNMP05
Password for 'https://ALNMP05@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/ALNMP05/Assessment05.git/'
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git push -u origin main
Enumerating objects: 56, done.
Counting objects: 100% (56/56), done.
Delta compression using up to 8 threads
Compressing objects: 100% (38/38), done.
Writing objects: 100% (56/56), 15.77 KiB | 2.25 MiB/s, done.
Total 56 (delta 5), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (5/5), done.
To github.com:ALNMP05/Assessment05.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$
```



```
narayanan@CYBERPUNK: ~/NetBeansProjects/Assessment05$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git log
commit 7aa6e91995c404f1958469f3356ff6e324dd936d (HEAD -> main, origin/main)
Author: lakshminarayanan <alnmp@gmail.com>
Date: Thu Oct 16 19:41:52 2025 +0530

    initial commit
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$
```

Creating develop branch from main, for development

```
Author: lakshminarayanan <alnmp@gmail.com>
Date: Thu Oct 16 19:41:52 2025 +0530

    initial commit
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git branch develop
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git checkout develop
Switched to branch 'develop'
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git add .
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git commit -m "created develop branch"
On branch develop
nothing to commit, working tree clean
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$ git push origin develop
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/ALNMP05/Assessment05/pull/new/develop
remote:
To github.com:ALNMP05/Assessment05.git
 * [new branch]    develop -> develop
narayanan@CYBERPUNK:~/NetBeansProjects/Assessment05$
```

We can add team members as collaborators in github, which gives them push access for the development.

Each members can create their own branch from develop branch and start development.

